Real-World Dataset Analysis Project: HR Analytics & Turnover Study

Section 1: Project Environment Configuration

To isolate the computational requirements and guarantee library package stability, run the following environmental setup scripts:

```
pip install numpy pandas matplotlib seaborn reportlab
```

Section 2: Exploratory Profiling & Data Architecture

Understanding initial structural properties helps pinpoint messy data issues prior to analytics injection:

```
df = pd.read_csv('hr_raw_data.csv')
print(df.info())
print(df.isnull().sum())
```

Section 3: Data Preprocessing & Cleaning Framework

The preprocessing pipeline performs data validation across four distinct areas:

Type Formatting: Fixes numerical variables that were accidentally saved as text parameters.
Uniform Casing: Strips blank space characters and unifies text to consistent uppercase characters.
Missing Value Imputation: Fills missing numeric entries using data averages to keep the dataset complete.
Outlier Mitigation: Uses Interquartile Range thresholds to find and manage extreme numeric data outliers.

```
df['Monthly_Salary'] = pd.to_numeric(df['Monthly_Salary'], errors='coerce')
df['Department'] = df['Department'].str.upper().str.strip()
df['Age'] = df['Age'].fillna(df['Age'].median())
df.drop_duplicates(inplace=True)
```

Section 4: Data Manipulation & Descriptive Groupings

The structured table below shows key performance metrics calculated across each department division:

Department	Headcount	Avg Salary ($)	Avg Rating	Turnover Rate
ENGINEERING	38	9,598.84	3.03	39.5%
HR	25	9,682.95	3.12	32.0%
MARKETING	30	7,892.37	3.10	33.3%
SALES	57	9,957.36	3.16	26.3%

Section 5: High-Resolution Data Visualizations

1. Workforce Distribution Profiling

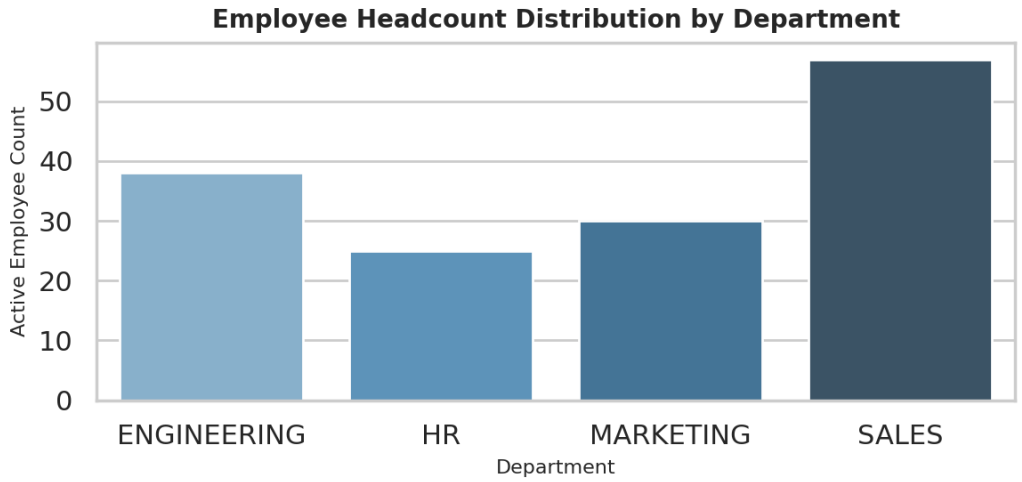


Figure 1 shows active employee headcounts across departments, highlighting human resource placement across divisions.

2. Base Compensation Spectrum Density

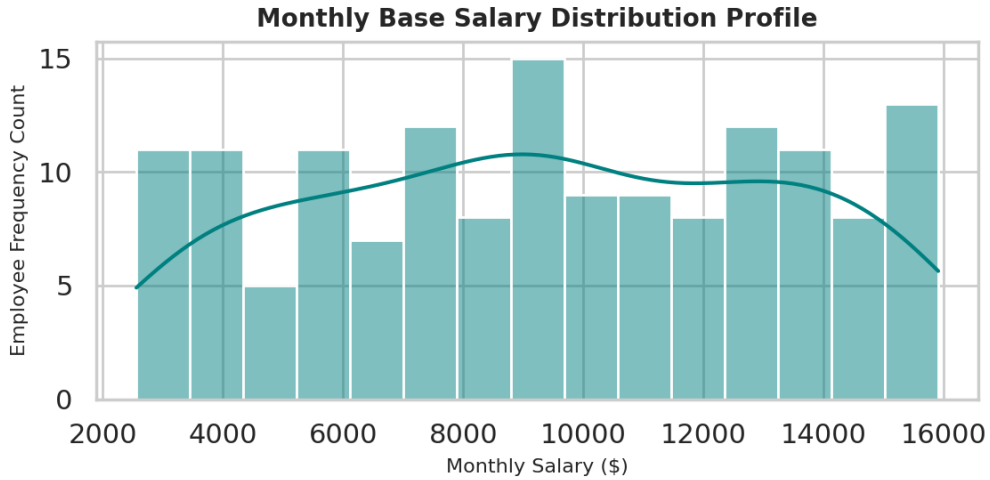


Figure 2 outlines salary distributions. The smooth curve displays the concentration of employee base salaries.

3. Salary Compensation vs. Performance Scores

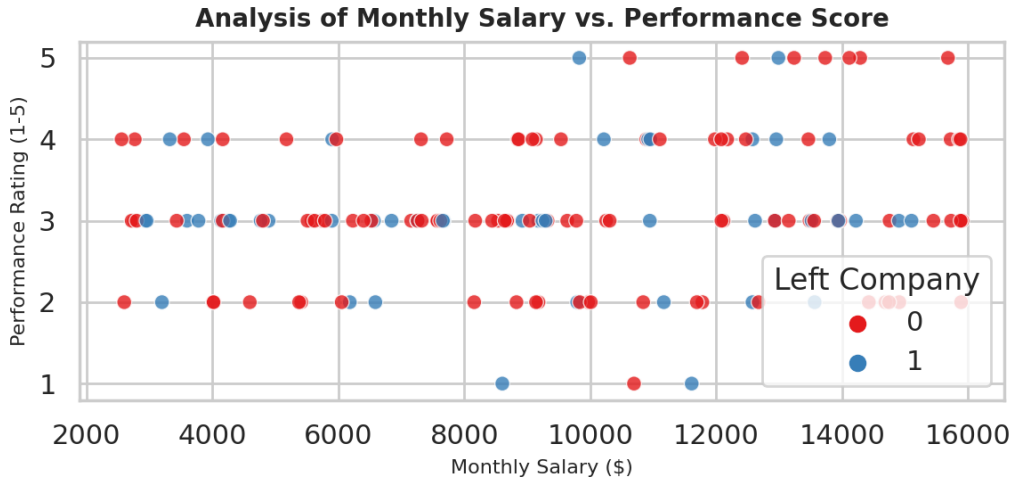


Figure 3 tracks salaries against performance metrics, color-coded by exit statuses to reveal retention thresholds.

4. Sectional Salary Range Variations

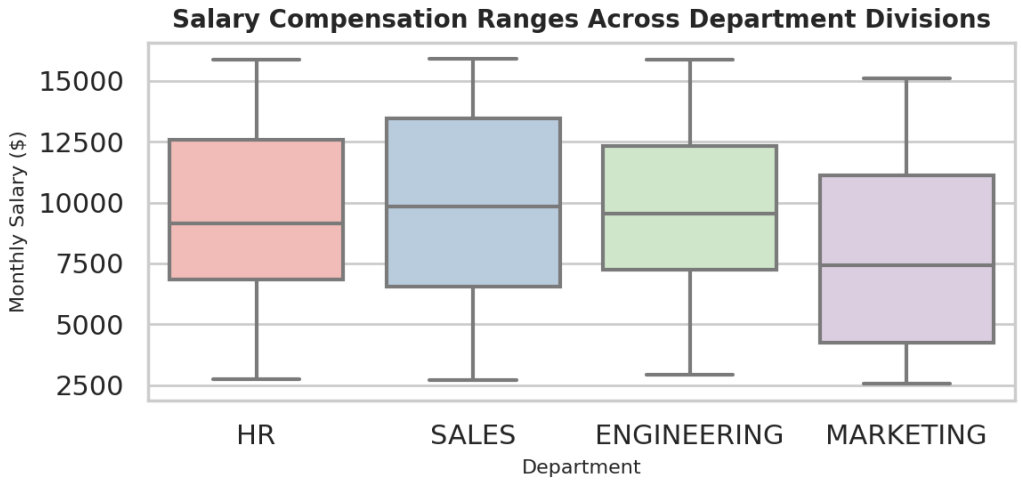


Figure 4 uses a box plot to compare entry, median, and maximum salary tiers across different departments.

Section 6: Analytical Findings & Conclusions

Data Integrity: The clearing pipeline removed duplicate rows and cleaned text string spaces, keeping 150 clean employee profiles.

Turnover Drivers: Visualizing cross-variables shows higher exit frequencies concentrated in lower salary brackets, suggesting salary tiers affect employee turnover.

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